

Sources — Wildfire Impact on Timber Supply Chains & CO2 Emissions

Supply Chain / Lumber Price Data

- **2017 Tubbs Fire** — lumber prices surged ~50% within a few months following the fire, which destroyed over 5,600 buildings (industry/lumber market reporting)
- **California Air Resources Board (CARB) & general market reporting** — California supplies roughly 10% of national lumber; approximately 60% of the state's valuable forests sit in high-wildfire-risk zones
- **2025 Eaton/Palisades wildfires impact projections** — analysts projected ~6% California lumber cost increase within 6 months, and ~3.5% national increase in 3–6 months, rising to ~15% over 15 months
- **Trade/industry data** — Canada supplies approximately 80% of U.S. softwood lumber imports
- **2023 Canadian wildfire season reporting** — Alberta and Quebec accounted for 44% of Canadian lumber shipments and 45% of Q1 2023 production; Resolute Forest Products temporarily closed four Quebec sawmills
- **Canadian construction cost data** — construction costs were 7.5% higher in June 2023 versus the prior year
- **Lumber futures market reporting** — a lumber futures contract jumped more than 10% in a single day, triggering trading circuit breakers in 2023; Canfor projected a 115-million-board-foot production loss; CN/CP faced rail disruption near Vancouver
- **Southeast U.S. wildfire/pulpwood market reporting** — pine pulpwood prices fell approximately 60% (April to May) due to a salvage-wood glut following fires across Florida and Georgia

CO2 Emissions Data

- **California Air Resources Board (CARB)** — Wildfire Emission Estimates for 2020 4.2 million acres burned produced 106.7 million metric tons of CO2
- **California Air Resources Board (CARB)** — Wildfire Emission Estimates 2000–2021 2021 data (2.5M acres → 85.2 MMT CO2); Dixie Fire alone contributed 37.4 MMT CO2
- **Nature** — Carbon emissions from the 2023 Canadian wildfires Satellite-based inverse modeling estimating 647 TgC released May–September 2023, comparable to a top-10 national emitter
- **World Resources Institute / Global Forest Watch** — Canada's Record-Breaking 2023 Wildfires Released Nearly 4 Times More Carbon than Global Aviation 7.8 million

hectares burned produced roughly 3 billion tons of CO₂ — nearly 4x global aviation's 2022 emissions

- **Copernicus Atmosphere Monitoring Service (CAMS)** — Canada produced 23% of global wildfire carbon emissions for 2023 ~480 megatonnes of carbon from Canada in 2023, about 5x the 20-year average, 23% of the global wildfire carbon emissions total that year
- **U.S. EPA** — Technical Methods for Wild and Prescribed Fires Fuel-loading consumption caps of 5 tons/acre in the Eastern U.S. versus 20 tons/acre in the West — basis for the regional (Southeast vs. West) CO₂ emission-factor split used in the simulator
- **PMC / NIH** — Carbon emissions from the 2023 Canadian wildfires Companion paper to the Nature study, with additional National GHG Inventory Report (NGHGI) and forest carbon detail
- **NASA Jet Propulsion Laboratory (JPL)** — New NASA Study Tallies Carbon Emissions From Massive Canadian Fires Corroborating estimate of approximately 640 million metric tons of carbon released in 2023
- **440 Megatonnes / Canadian Climate Institute** — How forest fires affect Canada's climate progress Canada's National Inventory Report figure of nearly 1 gigatonne (1 billion tonnes) of CO₂ released from Canada's forests in 2023

Compiled for the wildfire supply chain research and the `wildfire_supply_chain_simulator.py` Streamlit app. Figures are drawn from cited third-party sources; the simulator itself is an illustrative model, not a validated forecast.